

MEMVERSE — Zero-Trust Privacy Gateway for AI Memory

Solving the Personalization vs. Privacy Paradox for Text, Documents & Multimodal AI

📌 Project Description (2-3 Line Summary):

MEMVERSE is a client-governed zero-trust privacy gateway that empowers AI assistants to deliver highly personalized responses without exposing raw personal identifiers to external LLMs. Through an automated 12-stage security pipeline, dynamic policy transformation (names to roles, exact numbers to bands, locations to regions), cryptographic Memory Passports, and tamper-evident SHA-256 ledger receipts, MEMVERSE guarantees mathematical privacy for text, PDF documents, and biometric images.

👥 Team Information

Role	Name	Institution & Details
Team Lead & Developer	Satvik Kesarwani	Indian Institute of Information Technology (IIIT), Pune B.Tech — Computer Science & Engineering (MIS: 112315166)
Team Name	Team MEMVERSE (Zero-Trust AI Privacy Initiative)	

🔗 Project Links & Live Deployments

🌐 Live Web Application (Vercel)

Interactive React/Vite app with ChatView, Identity Vault & Trace Radar:
<https://memverse-satvikkesarwanis-projects.vercel.app>

⚙️ Gateway API Service (Render)

FastAPI zero-trust gateway with live NVIDIA NIM integration:
<https://memverse-api.onrender.com/api/status>

📁 Source Code Repository (GitHub)

Monorepo including frontend, backend, test suite & Playwright E2E:
<https://github.com/satvikkesarwani/memverse>

📁 Memverse Project Details (Drive)

Submission folder containing video demo, architectural assets & documentation:
https://drive.google.com/drive/folders/1U03jMmCsiOP9CiwQckT6S5NyeJYVo_f9?usp=sharing

⚡ Core Innovations & Technical Highlights

1. 12-Stage Zero-Trust Pipeline: Synchronous pipeline performing NER detection, prompt injection defense, policy transformation, and egress gatekeeping in <15ms.

2. Multimodal Privacy Engine: Automated redaction for uploaded PDF grade sheets and browser-native facial biometric detection with explicit consent.

3. Cryptographic Memory Passports: Scoped access tokens with strict TTL expiration and instant revocation mechanisms.

4. Tamper-Evident Hash Receipts: SHA-256 Merkle-style audit ledger providing non-repudiation and verifiable mathematical proofs.

Tech Stack: React 18, Vite, FastAPI, Python 3.11, NVIDIA NIM API, Cryptography (AES-256), face-api.js, PyPDF

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